

Discussion 1a Problems

Problem 1: Propositional Logic Language

For each sentence, we write the proposition in formal logic and then its negation.

Part (a): The cube of a negative integer is negative.

Proposition:

$$\forall z \in \mathbb{Z}, \quad z < 0 \Rightarrow z^3 < 0$$

Negation:

$$\exists z \in \mathbb{Z}, \quad z < 0 \wedge z^3 \geq 0$$

Part (b): There are no integer solutions to $x^2 - y^2 = 10$.

Proposition:

$$\nexists x \in \mathbb{Z}, \nexists y \in \mathbb{Z} \quad \text{s.t.} \quad x^2 - y^2 = 10$$

Negation:

$$\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z} \quad \text{s.t.} \quad x^2 - y^2 = 10$$

Part (c): There is one and only one real solution to $x^3 + x + 1 = 0$.

Proposition:

$$\exists x \in \mathbb{R} \text{ s.t. } x^3 + x + 1 = 0, \quad \wedge \quad \forall y \in \mathbb{R}, y^3 + y + 1 = 0 \Rightarrow y = x$$

Negation:

$$\forall x \in \mathbb{R}, x^3 + x + 1 \neq 0 \quad \vee \quad \exists y \in \mathbb{R} \text{ s.t. } y^3 + y + 1 = 0 \wedge y \neq x$$

Part (d): For any two distinct real numbers, we can find a rational number in between them.

Proposition:

$$\forall x, y \in \mathbb{R}, \quad x \neq y \Rightarrow \exists z \in \mathbb{Q} \text{ s.t. } x < z < y$$

Negation:

$$\exists x, y \in \mathbb{R}, \quad x \neq y \wedge \nexists z \in \mathbb{Q} \text{ s.t. } x < z < y$$

Problem 2: Order of Quantifiers

We determine which implications are always true regardless of $P(x, y)$, and give counterexamples for false ones.

(a) $\forall x, \forall y, P(x, y) \Rightarrow \forall y, \forall x, P(x, y)$

True. Swapping the order of two universal quantifiers does not change the meaning of the statement.

(b) $\exists x, \exists y, P(x, y) \Rightarrow \exists y, \exists x, P(x, y)$

True. Swapping the order of two existential quantifiers does not change the meaning of the statement.

(c) $\forall x, \exists y, P(x, y) \Rightarrow \exists y, \forall x, P(x, y)$

False. The left side says: for each x , we can find some y (possibly depending on x). The right side says: there is a single y that works for every x simultaneously. These are not equivalent.

Counterexample: Let $P(x, y)$ be $y = x$ over $\mathbb{R}$. Then $\forall x, \exists y, y = x$ is true (just pick $y = x$). But $\exists y, \forall x, y = x$ is false, since no single real number equals every real number simultaneously.

(d) $\exists x, \forall y, P(x, y) \Rightarrow \forall y, \exists x, P(x, y)$

True. If there exists some x^* such that $P(x^*, y)$ holds for all y , then in particular for any given y , we can exhibit $x = x^*$ as a witness. So the right side holds.

Problem 3: Lewis Carroll's Symbolic Logic

We use the following predicates over the universe of all people:

- $P(x)$: x goes to a party
- $B(x)$: x has brushed his or her hair
- $F(x)$: x looks fascinating
- $U(x)$: x is untidy
- $O(x)$: x is an opium-eater
- $N(x)$: x has no self-command
- $K(x)$: x wears kid gloves

Part (a): Translating the six sentences

(I) No one going to a party fails to brush his or her hair:

$$\forall x, P(x) \Rightarrow B(x)$$

(II) No one looks fascinating if untidy:

$$\forall x, U(x) \Rightarrow \neg F(x)$$

(III) Opium-eaters have no self-command:

$$\forall x, O(x) \Rightarrow N(x)$$

(IV) Everyone who has brushed their hair looks fascinating:

$$\forall x, B(x) \Rightarrow F(x)$$

(V) No one wears kid gloves unless going to a party:

$$\forall x, K(x) \Rightarrow P(x)$$

(VI) A person is always untidy if they have no self-command:

$$\forall x, N(x) \Rightarrow U(x)$$

Part (b): Contrapositives

(I) $\forall x, \neg B(x) \Rightarrow \neg P(x)$

(II) $\forall x, F(x) \Rightarrow \neg U(x)$

(III) $\forall x, \neg N(x) \Rightarrow \neg O(x)$

(IV) $\forall x, \neg F(x) \Rightarrow \neg B(x)$

(V) $\forall x, \neg P(x) \Rightarrow \neg K(x)$

(VI) $\forall x, \neg U(x) \Rightarrow \neg N(x)$

Part (c): What can we conclude about a person who wears kid gloves?

Starting from $K(x)$, we chain the propositions as follows:

$K(x) \Rightarrow P(x)$	(V)
$\Rightarrow B(x)$	(I)
$\Rightarrow F(x)$	(IV)
$\Rightarrow \neg U(x)$	(II, contrapositive)
$\Rightarrow \neg N(x)$	(VI, contrapositive)
$\Rightarrow \neg O(x)$	(III, contrapositive)

Conclusion:

$$\forall x, K(x) \Rightarrow \neg O(x)$$

Anyone who wears kid gloves is not an opium-eater.